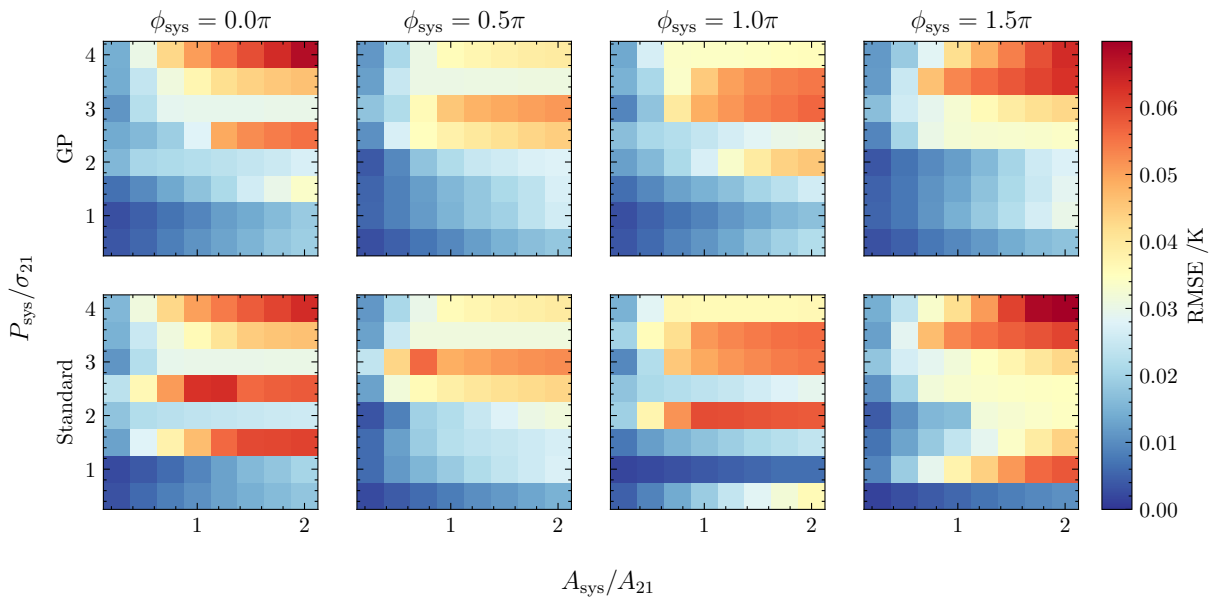

Figure 6. Weighted root mean square error (equation 18) of the pipeline fits for the standard pipeline (lower row) and the Gaussian processes pipeline (upper row) for different systematic parameters.

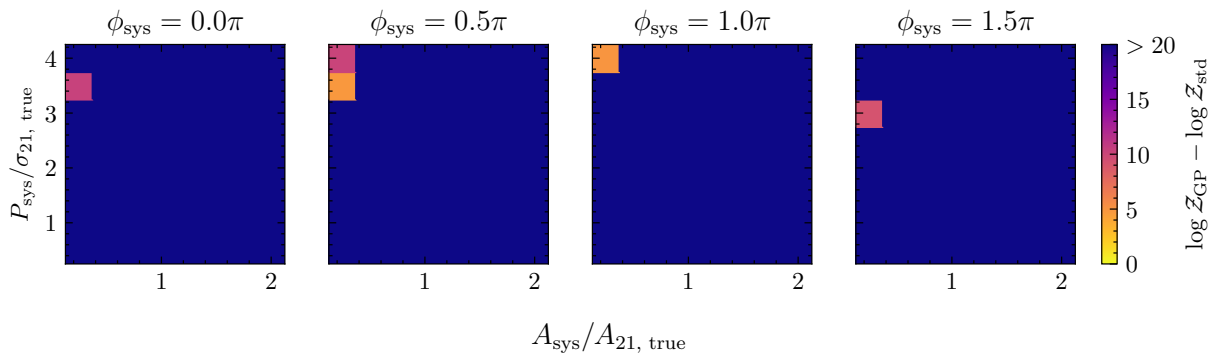

Figure 7. Log Bayes factor (equation 19) comparing the Bayesian evidence of the pipeline fits for the standard pipeline, $\mathcal{Z}_{\rm std}$ and the Gaussian processes pipeline, $\mathcal{Z}_{\rm GP}$, for different systematic parameters.

the data to within 2σ by the Gaussian process pipeline. This shows that by taking into account the time correlation of the systematics, the Gaussian pipeline is less likely to fit a trough of the sinusoidal systematic.

3.3 Gaussian Process Regression

Gaussian processes also enable investigation of systematic structure over time by allowing the calculation of a regression line for the model residuals. Using equations 16 and 17 we get a mean temperature of the model residuals with time, indicating trends in the amplitude of the systematic.

Figure 13 shows the Gaussian process time regression for the 85 MHz frequency bin for data with 24 time bins of length 15 minutes. A foreground-modulated systematic with an initial amplitude of $A_{\rm sys}$ = 0.209 K was added to the data. The black points are the residual temperatures after the mean foreground and signal models had been subtracted. The red dot-dashed line in the true signal amplitude and given by equation 13. The blue line is the GP regression line determined using equation 16 and the blue shaded area is the $\pm 1\sigma$ error determined using equation 17.

While the regression line doesn't capture the same shape of the true systematic, the true line is within error of the regression and the GP captures the behaviour that the systematic is increasing with time. As the data is noisy it is unlikely that the GP will be able to capture the exact details of the systematic modulation.

The method outlined in this paper is general and only assumes that the systematic varies with time, we can use the GP regression line to infer how we would expect the systematic to change with time. In future work where we may model and fit for time-varying systematics, continuing on from the time-averaged work of Scheutwinkel et al. (2022), the GP regression is a useful base to inform the choice of model.

4 CONCLUSIONS

In this paper we presented a new method of mitigating the effects of unmodelled systematics when fitting for the global 21-cm signal in radio cosmology experiment data. By using a squared exponential

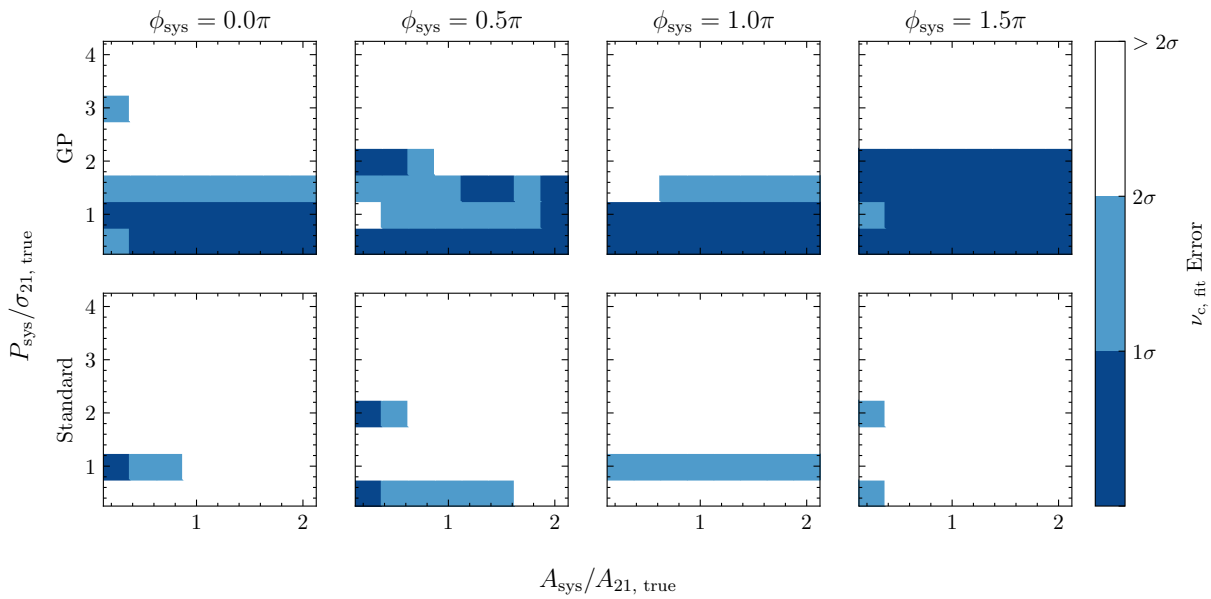

Figure 8. Error in the fitted value of the centre frequency of the Gaussian signal model for the standard pipeline (lower row) and the Gaussian processes pipeline (upper row) for different systematic parameters.

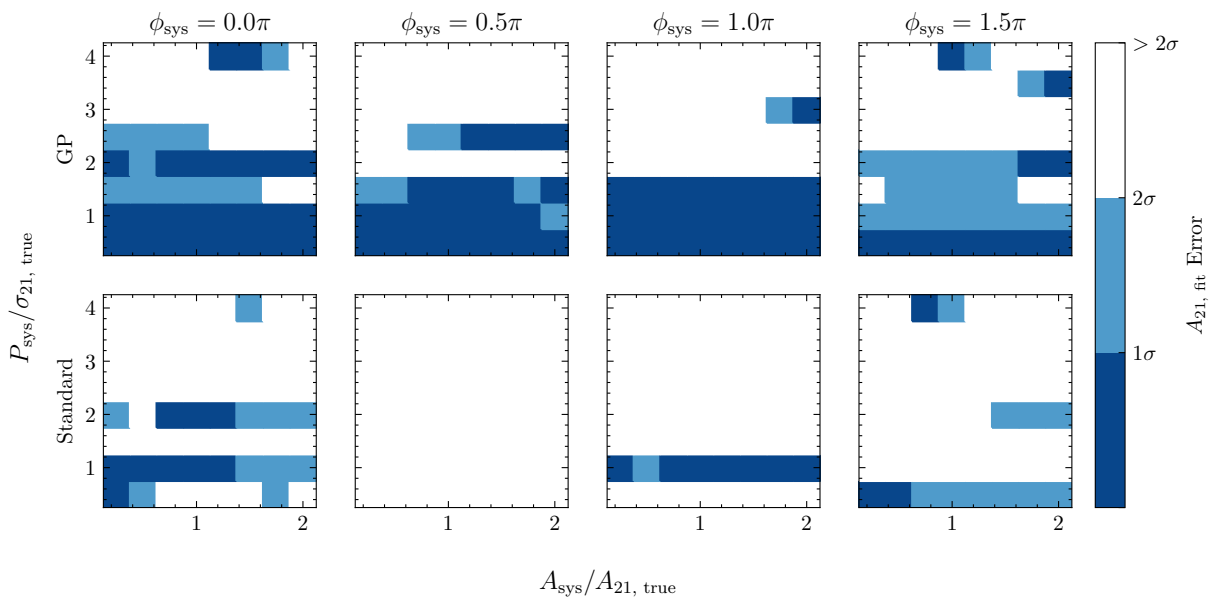

Figure 9. Error in the fitted value of the amplitude of the Gaussian signal model for the standard pipeline (lower row) and the Gaussian processes pipeline (upper row) for different systematic parameters.

Gaussian process kernel to fit for the correlations between time bins in the model residuals we are able to identify and mitigate the effects of time-varying residual systematics in the data.

We found that the Gaussian process pipeline was able to account for the presence of residual systematics in the data by widening the signal posteriors, reflecting our increased uncertainty in the signal parameters. We can use the squared-exponential kernel scale, $\sigma_{\rm SE}$, to identify the presence of systematics in the data as its value will be non-zero should there be a systematic. This demonstrates the method's power as a diagnostic tool.

Furthermore we saw that the Gaussian process pipeline generally improved the goodness-of-fit, as measured using a root mean square value, with a 5% improvement in RMSE values on average for the systematics we tested. Comparing the fitted signal parameters with the true values demonstrated that the GP pipeline reduces the biasing for systematics with a period less than twice the signal width. In many cases we found that the GP pipeline can recover parameters to within 1σ despite the standard pipeline parameter fits being further than 2σ from the true value. This can mainly be attributed to the widening of the posterior.

Further work will including using the Bayesian evidence to compare Gaussian Process kernels, as comparisons to other kernels could be beneficial to our understanding of the systematics. For example, a periodic kernel (Rasmussen & Williams 2006) could be useful when

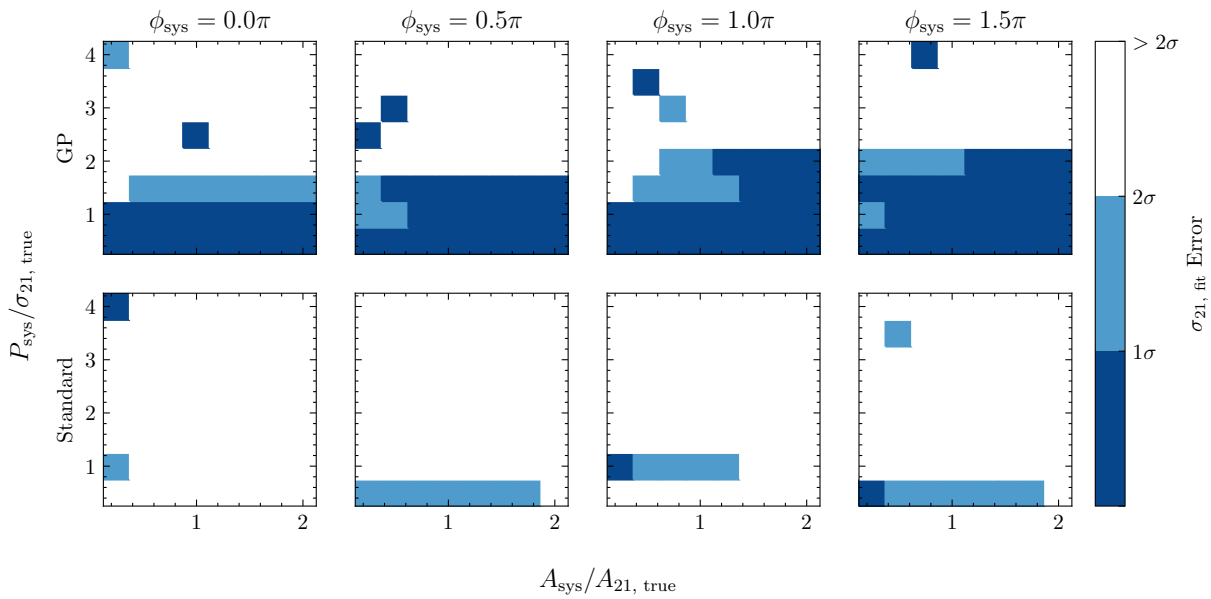

Figure 10. Error in the fitted value of the full-width half maximum of the Gaussian signal model for the standard pipeline (lower row) and the Gaussian processes pipeline (upper row) for different systematic parameters.

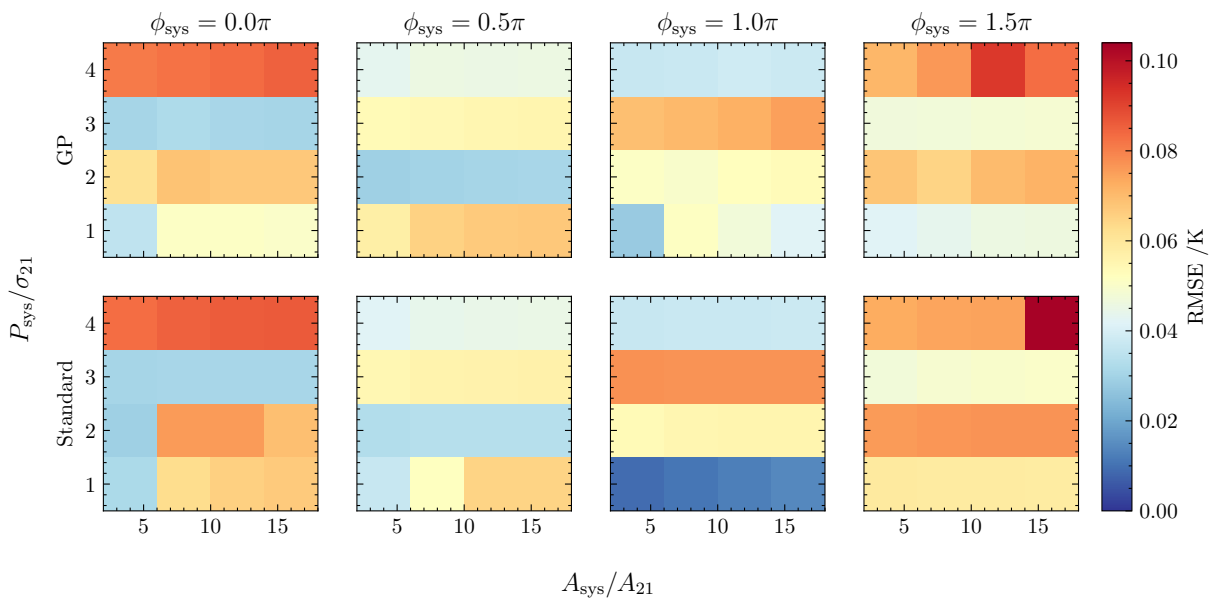

Figure 11. Weighted root mean square error (equation 18) of the pipeline fits for the standard pipeline (lower row) and the Gaussian processes pipeline (upper row) for systematics with amplitudes which greatly exceed that of the injected signal amplitude.

the systematic is modulated by the galactic foreground power as it is expected to vary periodically on a 24 hour timescale. A 2D Gaussian process could also be used to introduce correlations between frequency bins as well as the time bins.

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DATA AVAILABILITY

The data that supported the findings of this article will be shared on reasonable request to the corresponding author.